



NVIDIA Regional R&D Hub — Cost Breakdown

Leaf-Spine + Star · 2 Spines + 3 Leaves (Traffic-Function)

Leaf-1 COMPUTE · Leaf-2 SERVICES · Leaf-3 EDGE · 9 VLANs · OSPF · PT-compatible

All prices EUR, indicative (June 2026)

1. Leaf-Spine Core Fabric

2 Spines + 3 Leaves = 8 physical switches + 4 StackWise cables. In Packet Tracer simulation, this is modelled with 5 switches for leaf and spine

Item	Model	Qty	New/unit	New total	Refurb/unit	Refurb total
Spine Switch	Cisco C9300-24T-E	2	€2,433.97	€4,867.94	€562.94	€1,125.88
Leaf-1 COMPUTE	Cisco C9300-24T-E	2	€2,433.97	€4,867.94	€562.94	€1,125.88
Leaf-2 SERVICES	Cisco C9300-24T-E	2	€2,433.97	€4,867.94	€562.94	€1,125.88
Leaf-3 EDGE	Cisco C9300-24T-E	2	€2,433.97	€4,867.94	€562.94	€1,125.88
StackWise cables (per pair, prod)	Cisco STACK-T1	4	€261	€1,044	€46	€184
TOTAL		12		€20,515.76		€4687.52

2. Edge & Access Devices

Identical for all department access — 6 access switches (one per dept: Production, Management, Study, Support-A, Support-B, IT).

Item	Model	Qty	New/unit	New total	Refurb/unit	Refurb total
Edge Router (x2)	Cisco ISR4331	2	€453	€906	€70	€140
Firewall	Cisco firepower 1120	1	€2,095.50	€2,095.50	€983	€983

Access SW (per dept)	Cisco C9200-48T-E	6	€2,142	€12,852	€1,428	€8,568
TOTAL		9		€15,853.50		€9,691

3. Servers — VLAN 70 (Services) + DMZ

All hang off Leaf-2 (Services). DMZ Web/FTP server sits behind Leaf-3 (Edge) on VLAN 80. iSCSI restricted to Production VLAN 10 via ACL.

Item	Model	Qty	New/unit	New total	Refurb/unit	Refurb total
DNS Srv	Dell R450	1	€3,266	€3,266	€3,266	€3,266
DHCP Srv	Dell R450	1	€3,266	€3,266	€3,266	€3,266
AAA/RADIUS Svr	Dell R450	1	€3,266	€3,266	€3,266	€3,266
Syslog Svr	Dell R450	1	€3,266	€3,266	€3,266	€3,266
HTTP Svr	Dell Power edge R550 8SFF	1	€2,932	€2,932	€2,932	€2,932
iSCSI Storage Svr	Dell R750 NVMe	1	€6,503	€6,503	€3,074	€3,074
Compute Server	Dell R750 GPU	1	€10,800	€10,800	€5,232	€5,232
DMZ FTP Svr	Dell Power edge R550 8SFF	1	€2,932	€2,932	€2,932	€2,932
Internal FTP Svr	Dell Power edge R550 8SFF	1	€2,932	€2,932	€2,932	€2,932
TOTAL		9		€39,163		€30,166

4. IAM Workstations — 48 total across 6 departments

Item	Model	Qty	New/unit	New total	Refurb/unit	Refurb total
IAM WS Production	GPX-WS 740 RTP	10	€7,499	€74,990	€1,380 (approx)	€13,800
IAM WS Study	HP Z4 G4 + Windows 11	8	€1,095	€8,760	€874	€6,992
MIA WS Mgmt	HP Z4 G4 + Windows 11	5	€1,095	€5,475	€874	€4,370

IAM WS Supp-A	HP Z4 G4 + Windows 11	10	€1,095	€10,950	€874	€8,740
IAM WS Supp-B	HP Z4 G4 + Windows 11	10	€1,095	€10,950	€874	€8,740
IAM WS IT	HP Z4 G4 + Windows 11	5	€1,095	€5,475	€874	€4,370
Monitors	HP series 5 Pro 524pf	38	€163	€6,194	€129	€4,902
Printer	HP Color LaserJet Enterprise MFP	4	€404	€1,616	€404	€1,616
TOTAL		90		€1,24,410		€53,530

TOTAL HARDWARE (Core + Edge + Servers + Workstations)	€199,942.26		€98,074,52
--	--------------------	--	-------------------

5. Wiring & Labour

Wiring & Infrastructure

Item	New	Refurb/Reuse
CAT6A patch cables x100	€180	€180
CAT6A 1m cables x50	€120	€120
OM3 fibre duplex x20 (spine-leaf)	€280	€280
OM3 fibre 5m x10	€230	€230
SFP+ 10G transceivers x24	€1104	€552
Serial DCE cables x2	€60	€30
19" Rack 42U x2	€920	€460
APC UPS 3000VA x2	€1840	€920
Patch panels 48p x3	€460	€230
Cable management	€460	€460

Labels + tools	€120	€120
TOTAL WIRING	€5,774	€3,582

6. Project Add-On Costs (Licensing, Support & Hardware Accessories)

Role	PT Simulation
Cisco SmartNet	€7,000
VMware licensing	€6,000
Windows Server	€9500
NVIDIA GPU licensing	€5,000
Rack PDUs	€1400
Consumables	€1200
TOTAL	30100

7 . Labour — 6-person team, 8-day project

Role	Day rate	Days × Cost
Project Fees		€5000
Project Manager	€600/d	6d = €3,600
Lead Architect	€600/d	5d = €3,000
Network Config Engineer	€600/d	6d = €3,600
Routing Config Engineer (HSRP+OSPF)	€600/d	5d = €3,000
Security Engineer	€800/d	5d = €4,000

Test Engineer	€850/d	4d = €3,400
Physical Installation	€500/d	3d = €1,500
Post-install support	€600/d	3d = €1,800
TOTAL LABOUR	€28900	

8. Total Cost of Project

Scenario	Hardware	Wiring	Labour	Add-ons	TOTAL (rounded)
NEW hardware	€199,941.76	€5,774	€28900	€30,100	€2,64,716
REFURBISHED hardware	€98074,52	€3,582	€28900	€30,100	€1,60,657

9. Insurance

Insurer	Annual Premium	Coverage Type	Coverage Amount	What it protects
AXA	€1,800 / year	Hardware & Electronics	€350,000 – €600,000	48 workstations, 10 high-end GPU systems, server room equipment, switches, MFPs, racks
Hiscox	€2,200 / year	Cyber Risk + E&O (Professional Liability)	€500,000 – €1,000,000	Cyber-attacks, ransomware, data breaches, AI/LLM development liability, IP disputes, professional mistakes
TOTAL	€4,000 / year €20,000 for 5 years		€850,000 – €1,600,000	Combined protection for hardware + cyber + liability

10. Final Pricing Table

Package	Solution	Equipment Cost	Insurance	Reserve	Total Project Cost (Excl. VAT)	Recommendation
Enterprise Package ★	New Products	€264,716	€20,000*	€10,000	€294,716	Recommended – Maximum reliability & protection
Standard Package	New Products	€264,716	€4,000**	€10,000	€278,716	Basic protection option
Value Package	Refurbished Products	€160,657	€20,000*	€10,000	€190,657	Budget-focused solution
Economy Package	Refurbished Products	€160,657	€4,000**	€10,000	€174,657	Lowest investment option

* for 5 years

** for 1 year

Note:

All costs are indicative and subject to change based on vendor pricing, licensing, and scope variations. Any approved changes or additions will first be covered by the project reserve; any excess costs beyond the reserve shall be borne by the Client, subject to prior written approval under the formal change control process.

11. Sources — Where Each Price Was Checked

Direct links to the listing used for each device's indicative price (June 2026). Prices fluctuate — re-check before final procurement.

Device / Cost Item	Source	Link
Cisco C9300-24T-E (Spine/Leaf, new)	it-planet — New	https://it-planet.com/nl/p/cisco-c9300-24t-e-381667.html?number=5645211000
Cisco C9300-24T-E (Spine/Leaf, refurb)	it-planet.com	https://it-planet.com/nl/p/cisco-c9300-24t-e-381667.html?number=5645211000.1&gad_source=&gad_campaignid=17415087807&gbraid=0AAAAAC89HVSwQ8eFTdRJKqANGoDBVMDQ9&gclid=CjwKCAjwgO7RBhBKEiwAZNP85ryTTq-GtVyW0FVppy4TAFofw-M3YcRDhp9FSixcgFiM_CqeGpqDBPhoCuTkQAvD_BwE
Cisco STACK-T1 (StackWise cable)	Bechtle.com	https://www.bechtle.com/nl/shop/cisco-catalyst-stack-t1-1m-cable--817441-03--p
Cisco ISR4331 (Edge Router)	Amazon.com.be	https://www.amazon.com.be/-/en/Cisco-Integrated-Services-Router-4331/dp/B00Q4E8OYK?language=en_GB
Cisco ISR4331	Amazon.be	https://www.amazon.com.be/-/en/Cisco-Firepower-Security-Firewall-Appliance/dp/B00UJV073AE
Cisco firepower 1120 (Firewall, new/refurbished))	Tonitrus	https://www.tonitrus.com/FPR1120-NGFW-K9_8?gad_campaignid=19965757386&gad_source=1&qgbraid=0AAAAAC8nqXXLrRbVvx6BBJ_ykDIM7L1t1h&qclid=CjwKCAjwgO7RBhBKEiwAZNP85o6CRvoTb2gLKra6HrsJ1ZjQioJwf4T5xXq_8mp3fRCUJnG82_ZD8SxoCERMQAvD_BwE
Cisco C9200-48T-E (Access Switch) new/refurbish	IT-panet	https://it-market.com/en/switches/gigabit/cisco/c9200-48t-e/753688-E2-165137?gad_source=1&gad_campaignid=22970621912&qgbraid=0AAAAADqe1y6f4GQ3X0pmOFGNFpiEQ_Tw4&qclid=Cj0KCQjwo_PRBhDNARIsAEcVALXaunbB6HTKVBPCuj93d4ggaqIT9A5sV9sW1LzX-nw1VSR7tAH0o1waAn7dEALw_wcB
Dell PowerEdge R450 (Servers)-new/refurbished	Server Mall online portal	https://servermall.com/sets/dell-r450-servers/?srsltid=AfmBOopCBXHtuoq83clry2XjV9OGkzV3QNZr4wBYgZkZTxiBeQSp7I3
Dell PowerEdge R550 (Servers) -new	Servermail online portal	https://servermall.com/catalog/servers/dell-poweredge-r550-16sff/?utm_source=google&utm_medium=cpc&utm_campaign=EU_GMC_Servers.18945982072&utm_content=.146588424914..615501702627&utm_term=&placement=.pla-293946777986&gad_source=1&gad_campaignid=18945982072&qgbraid=0AAAAAogUINVt-rUNFODc_Mw8UW9FNw4KP&qclid=Cj0KCQjwo_PRBhDNARIsAEcVALWDM_j87D-zIJfBPbkIk8ZLX9jIKRZNfiCTGe7RrL4oahaU-63V0oaApbOEALw_wcB

Dell PowerEdge R750 (Compute/iSCSI) -new	ETB-tech	https://www.etb-tech.com/dell-poweredge-r750xa-configure-to-order-btor750xa.html?srltid=AfmBOord7nvfVsiKuZoNq3ukljEBSIN5Mzabjdn3TBqXShiXomQlv7UX
Dell PowerEdge R750xa- refurbished	Bytestock	https://shop.bytestock.com/build-your-own-dell-r750xa-8-x-2-5-2u-rack-server
Dell PowerEdge R750 16SFF iSCI storage server- new	Server Schmiede	https://www.serverschmiede.com/konfigurator_bulk/en/dell-poweredge-r750-2u-server-24x-25-sff-16x-sas-8x-nvme-2x-intel-xeon-scalable-lga4189-ddr4-ecc-raid-2x-psu-1?gad_source=1&gad_campaignid=22024700381&gbraid=0AAAAADKZQnLXfKYVJFPKxnTxx_ZPh9gmk&gclid=CjwKCAjwgO7RBhBKEiwAZNP85mZx5uYoPfOgmZla-jCxeH5qPi38BtgcA9zE- k_BxuPtyfPwBW9WxoCrg4QAvD_BwE&cc=1782318325591
Dell PowerEdge R750 16SFF- refurbished	Servermall	https://servermall.com/catalog/servers/dell-poweredge-r750-16sff-refurbished/?utm_source=google&utm_medium=cpc&utm_campaign=EU_GMC_Servers.18945982072&utm_content=.146538424914..635501702627&utm_term=&placement=.pla-293946777986&gad_source=1&gad_campaignid=18945982072&gbraid=0AAAAAogUINVt-rUNFODc_Mw8UW9FNw4KP&gclid=Cj0KCQjwo_PRBhDNARIsAEcVALVJ_z0Nkd7OAZAcJgYTSzJcJiu3Qa_3_NiWnCNpjfl809_XfU4S93EaAonBEALw_wcB
Workstation for production GPX-WS 740RTP	Think mate website	https://www.thinkmate.com/system/gpx-ws-740rtp
HP Z4 G4 workstation - refurbished	refurbed	https://www.refurbed.be/p/hp-z4-g4/188307/?co=be&lang=nl&utm_cluster=gradeA&utm_content=hp-z4-g4-188307&utm_group=hp-z4-g4&ad_price=873.99&utm_campaign=BE%20-%20PMa%20-%20Under%20Index&utm_medium=cpc&utm_source=google&gad_source=1&gad_campaignid=21357765082&gbraid=0AAAAA9XhqOZql3c3L9dZYFij8M3TdubX7&gclid=Cj0KCQjwo_PRBhDNARIsAEcVALUI--34sQ23Epc7dq9q6_IK9hLfeawUBopWEuhNPRGYdsadrqHdcsaAgF0EALw_wcB
HP Z4 G4 workstation- new	Maas Computers	https://www.maascomputers.nl/home/3928-hp-z4-g4-workstation-intel-xeon-qc-w-2125-400-ghz-16gb-ddr4-512gb-ssd-quadro-p620-win-11-pro-0193808835636.html
Monitor- HP series 5 Pro 524pf	Cool blue	https://www.coolblue.be/nl/product/959957/hp-series-5-pro-524pf.html
Printers: HP Color LaserJet Enterprise MFP 430F	Dustin.be	https://www.dustin.be/nl/multifunctionals/hp/laserjet-enterprise-laserjet-enterprise-mfp-m430f-art-3pz55ab19-num-179114314/?utm_source=chatgpt.com
CAT6A cables, fibre, transceivers, racks, UPS	RS Online / FS.com / APC	https://www.fs.com · https://www.rs-online.com · https://www.apc.com
Labour rates (Belgium, network engineer)	ERI Economic Research Institute — salary data	https://www.ერი.com/salary/job/network-engineer/belgium
Insurance for Hardware	Axa- Full Machinery	https://www.axa.be/nl/ondernemers-kmo/gebouw-inhoud/verzekering-materieel
Insurance for Cybersecurity and E&O	Hiscox Cyber Insurance	https://www.hiscox.com/small-business-insurance/cyber-security-insurance

Note: The selected equipment is intended to represent enterprise-grade infrastructure suitable for a regional NVIDIA R&D hub. Exact models may vary during procurement; however, the chosen Cisco Catalyst, Cisco Firepower, Dell PowerEdge, and HP workstation platforms provide sufficient performance, scalability, and feature support for the proposed leaf-spine architecture, departmental segmentation, OSPF routing, VLAN-based isolation, and AI-focused compute workloads.

Note: The amount shown above may vary due to the fluctuation in rates on different sites shown above.